American Indian Casino Operations and Firm Long-Run Employment and Sales

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Main takeaway: Tribal casino openings generate persistent growth in reservation business activity, including among pre-existing firms.

Motivation: place-based development in Native communities

Policy context

- American Indian reservations face long-standing barriers to private-sector development.
- The Indian Gaming Regulatory Act (1988), or IGRA, created a legal framework for tribally owned casino operations.
- Tribal gaming became one of the largest sovereignty-driven economic development strategies in Native communities.

Tribal gaming gross gaming revenue

$43.9B

FY 2024

Open question

Did casino development translate into broader local business expansion inside reservation boundaries?

Research question and contribution

Research questions

1. Do tribal casino openings lead to sustained increases in reservation-based firm employment and sales?
2. Do these effects extend beyond the gaming industry, benefiting pre-existing firms and firms across other industries?

Contribution 1

New data link

Casino openings, establishments, and reservation boundaries

Contribution 2

Firm outcomes

Direct evidence on sales and employment

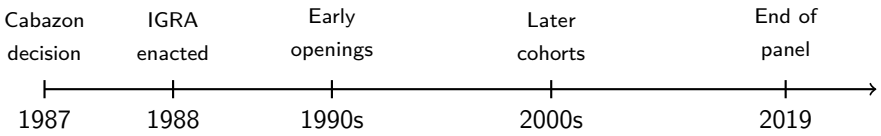
Contribution 3

Policy channel

Sovereignty-driven, place-based development and local spillovers

NETS = National Establishment Time Series.

Institutional setting: why casino openings are staggered



Key institutional features

- Casino openings required federal recognition, eligible trust land, and state–tribal compacts for Class III gaming.
- Opening years reflect legal, administrative, and compacting processes.
- IGRA did not allocate casino authorization through a federal need-based program.

Empirical implication

This setting creates staggered adoption across reservations and motivates a quasi-experimental interpretation of opening timing, subject to DiD assumptions.

Data construction

Inputs

- Tribal casino opening years, 1990–2019.
- NETS establishment-level panel data.
- Geocoded establishment locations.
- Census reservation boundary shapefiles.

Outcomes

- Reservation-year average establishment sales.
- Reservation-year average establishment employment.

Main samples

1. All establishments.
2. Pre-existing establishments.
3. All establishments excluding NAICS 71–72.

Main estimation sample

280 reservations

7,815 reservation-years

NAICS = North American Industry Classification System. NAICS 71–72 are arts, entertainment, recreation, accommodation, and food services.

Empirical design: staggered difference-in-differences

Estimator

We use the Callaway & Sant'Anna (2021) group-time average treatment effect framework for staggered adoption.

Compare the change in outcomes for reservations that open casinos in cohort (g) to the change in outcomes for never-treated reservations over the same period.

Treatment

- First year a tribal casino operates on reservation (r).
- Treatment remains on after opening.

Counterfactual group

- Never-treated reservations.
- Dynamic effects aggregated over event time.

ATT = average treatment effect on the treated.

Identification: what makes the comparison credible?

Identifying assumptions

- Parallel trends absent casino openings.
- No anticipation in measured outcomes.
- Stable never-treated comparison group.

Why plausible here

- Openings are shaped by sovereignty, federal recognition, land status, and compacting.
- IGRA did not allocate casino authorization based on economic need.

Diagnostics

- No systematic event-study pre-trends.
- Cohort raw trends track never-treated reservations before opening.
- Robustness to wild bootstrap inference.
- Robustness to short-gap, balanced-window, and covariate-adjusted specifications.
- Rambachan–Roth sensitivity checks.

Main results: average post-treatment effects

All establishments

Sales: +46.97%

Employment: +49.83%

Pre-existing

Sales: +16.63%

Employment: +26.12%

Excluding NAICS 71–72

Sales: +8.89%

Employment: +13.07%

Interpretation

The all-firm effect captures direct casino-related expansion and new activity. The pre-existing firm results show that growth is not purely mechanical: firms already operating before casino entry also expand.

Effects are transformed from log-point estimates using the standard exponential transformation.

Main results: Table 2

Table: CS Staggered Difference in Difference Regression Analysis for the Effects of Casino Openings on Reservation Level Sales and Employment

	(1) *ln*(Avg Sales)	(2) *ln*(Avg Emp)	(3) *ln*(Avg Sales)	(4) *ln*(Avg Emp)	(5) *ln*(Avg Sales)	(6) *ln*(Avg Emp)
Pre-Treatment						
Casino Effect	0.0673	-0.0202	0.0992	-0.0012	0.0534	0.0127
AT	(0.0676)	(0.0519)	(0.0663)	(0.0520)	(0.0646)	(0.0471)
Wild Bootstrap	(0.0653)	(0.0533)	(0.0705)	(0.0576)	(0.0642)	(0.0477)
Post-Treatment						
Casino Effect	0.3850	0.4041	0.1539	0.2321	0.0852	0.1227
AT	(0.0896)***	(0.0812)***	(0.0572)***	(0.0492)***	(0.0585)	(0.0481)**
Wild Bootstrap	(0.0878)***	(0.0784)***	(0.0544)**	(0.0495)***	(0.0602)	(0.0505)
Sample	All	All	Pre-Existing	Pre-Existing	Excl. 71–72	Excl. 71–72
Reservations	280	280	268	268	279	279
Percentage Effect	46.97%	49.83%	16.63%	26.12%	8.89%	13.07%
Mean of Outcome	1,310,281	19.89	1,313,116	19.90	770,153	10.11

How large are the effects?

All establishments

+47%
Sales

+50%
Employment

Pre-existing

+17%
Sales

+26%
Employment

Excl. NAICS 71–72

+9%
Sales

+13%
Employment

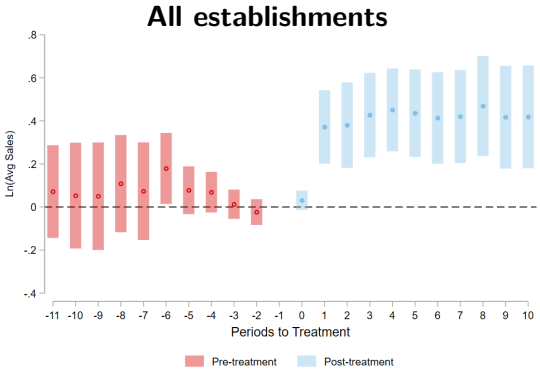
Magnitude in context

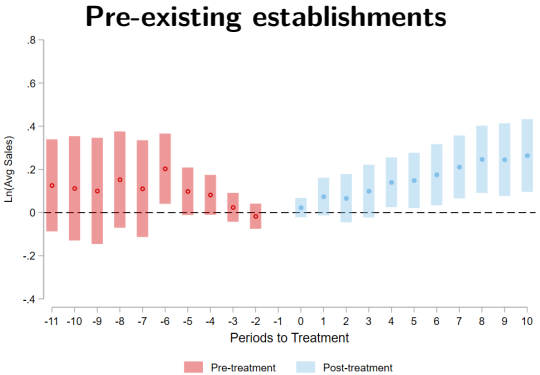
Prior tribal-gaming studies mainly estimate effects on household income, employment, health, education, or mobility. Our estimates measure a different margin: local private-sector business activity inside reservation boundaries.

Interpretation

The pre-existing-firm effects are the most informative for spillovers: they show that firms already operating before casino entry also expand after casino openings.

Dynamic sales effects

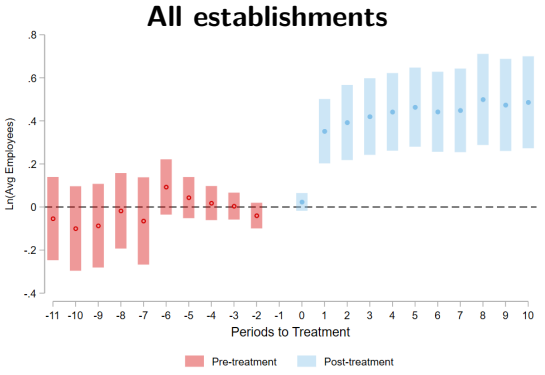


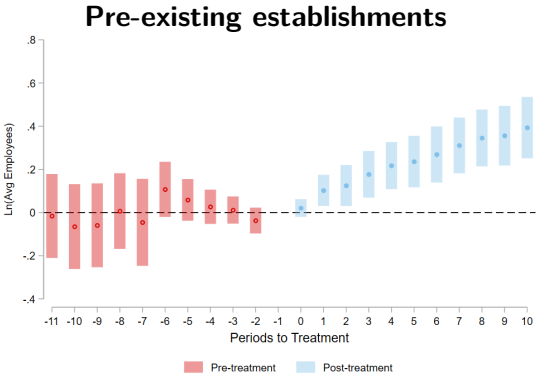


Takeaway

Sales rise immediately for all establishments and increase more gradually among pre-existing firms, consistent with spillovers to incumbent businesses.

Dynamic employment effects

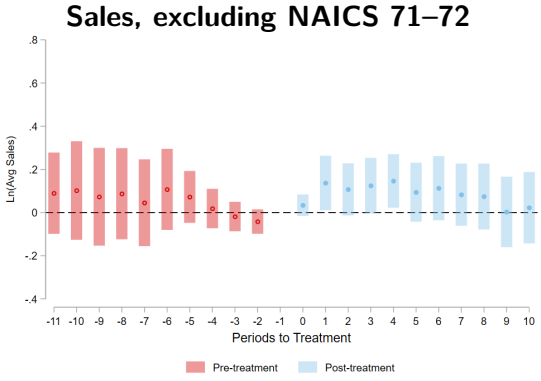


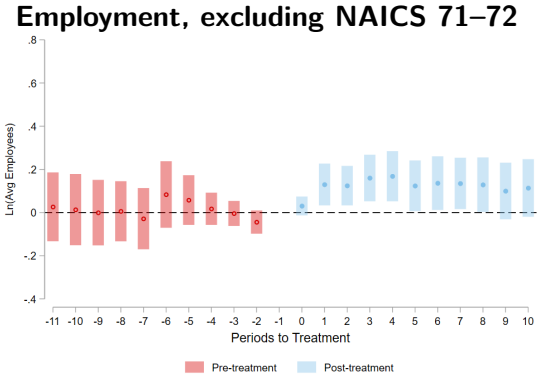


Takeaway

Employment effects are persistent and visible both in the full establishment sample and among firms operating before casino entry.

Are effects only gaming and hospitality?





Interpretation

The strongest effects appear in casino-adjacent and visitor-facing sectors. Outside NAICS 71–72, employment remains positive, while sales effects are smaller and less precisely estimated.

Policy implications

What the results suggest

- Tribal casino development can operate as a sovereignty-driven, place-based economic development strategy.
- Effects are not limited to the casino itself: pre-existing firms also expand.
- The strongest spillovers appear in visitor-facing and gaming-adjacent sectors.

What to be careful about

- Estimates capture reservation-level business responses to casino openings, not all tribal economic policies.
- Effects outside gaming and hospitality are more modest, especially for sales.
- Interpretation still depends on parallel trends and no anticipation.

Evaluations of tribal gaming should consider not only casino revenues or household outcomes, but also broader local business spillovers inside reservation economies.

What do the results imply?

Mechanisms consistent with the evidence

- Local demand from casino employees and visitors.
- Business-to-business spending.
- Increased visitor traffic to previously low-activity areas.
- Expansion of complementary hospitality and service sectors.

The evidence supports tribal casino development as a channel for broader reservation private-sector expansion.

Conclusion

Three takeaways

1. Tribal casino openings lead to large, persistent increases in reservation-level average establishment sales and employment.
2. Effects extend to pre-existing establishments, indicating expansion beyond new casino entry.
3. Spillovers appear strongest in gaming-adjacent and visitor-facing sectors, with more modest evidence outside NAICS 71–72.

Thank you.

Questions?

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Thank you.

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https://doi.org/10.3386/w32555

Backup slides

Backup: treatment timing

Cohort structure

- 27 treated cohorts in the analysis window.
- Casino openings are concentrated in the 1990s and early 2000s.
- 144 never-treated reservations in the cohort distribution table.

Why this matters

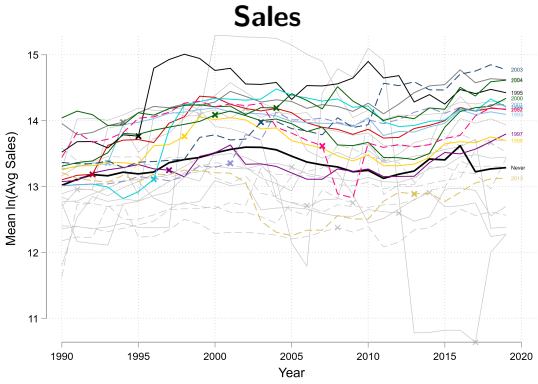
- Treatment timing varies across reservations.
- Treatment effects may be heterogeneous over cohorts and time.
- This motivates Callaway–Sant'Anna rather than a two-way fixed effects estimator.

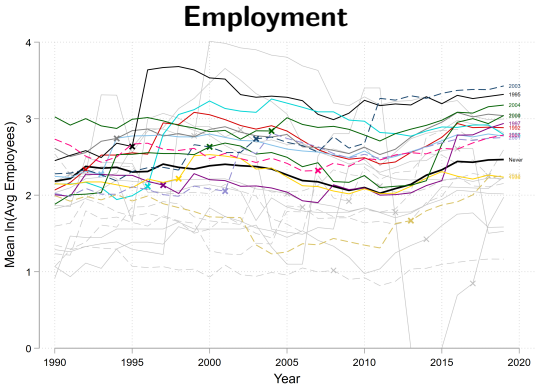
Analysis window

1990–2019

Treated and never-treated reservations

Backup: raw cohort trends



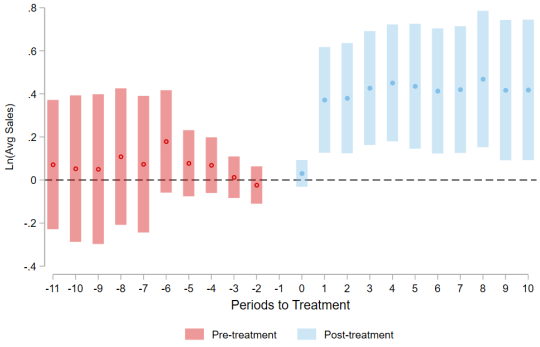


Diagnostic

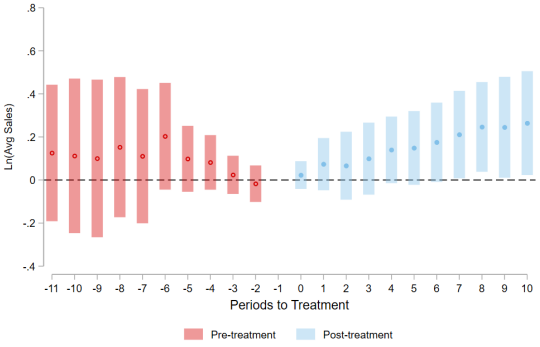
Pre-treatment trends for the highlighted treated cohorts closely track the never-treated group before casino openings.

Backup: wild bootstrap, sales

All establishments[0.2em]



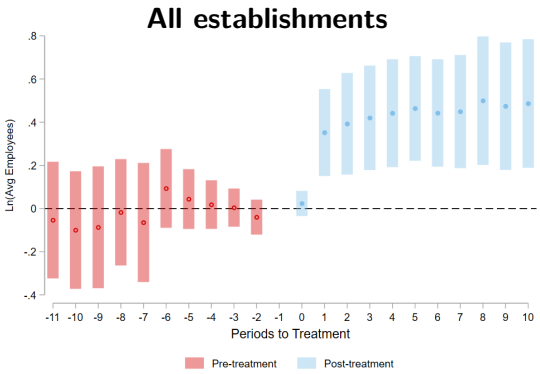
Pre-existing establishments[0.2em]

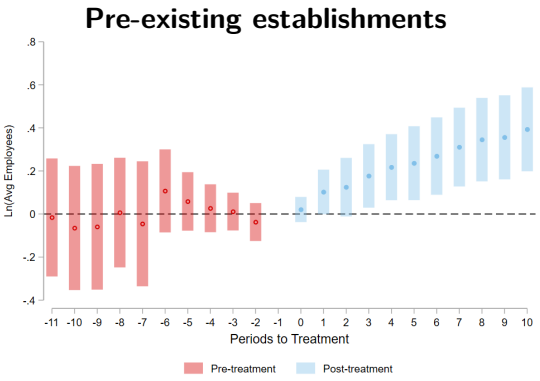


Robustness

The sales patterns remain similar using wild bootstrap confidence intervals.

Backup: wild bootstrap, employment

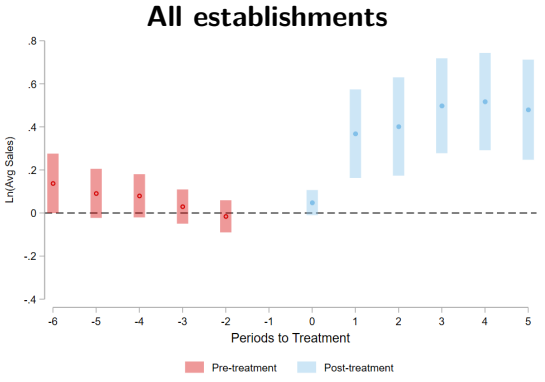


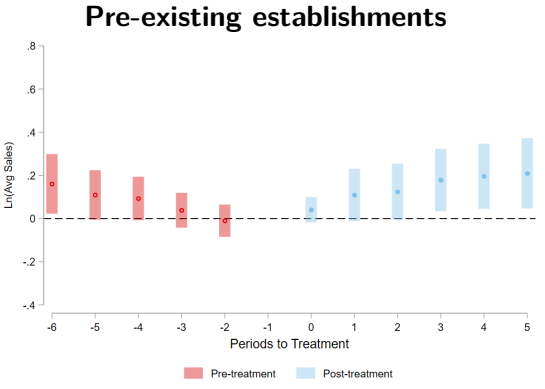


Robustness

The employment patterns remain similar using wild bootstrap confidence intervals.

Backup: balanced event-time window (-5,+5), sales

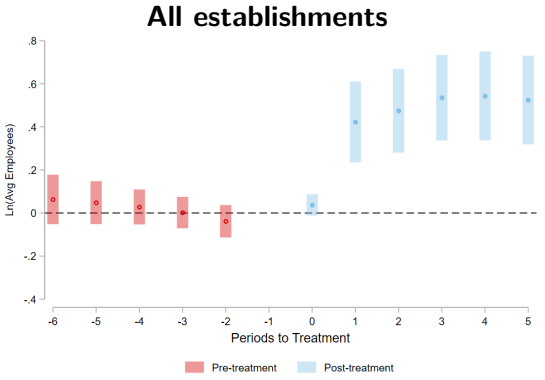


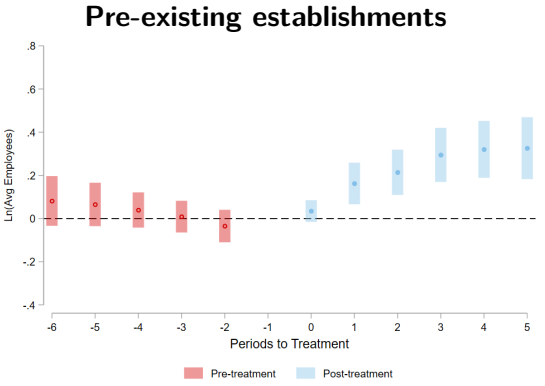


Robustness

Balanced event-time estimates preserve the main post-treatment sales pattern.

Backup: balanced event-time window (-5,+5), employment

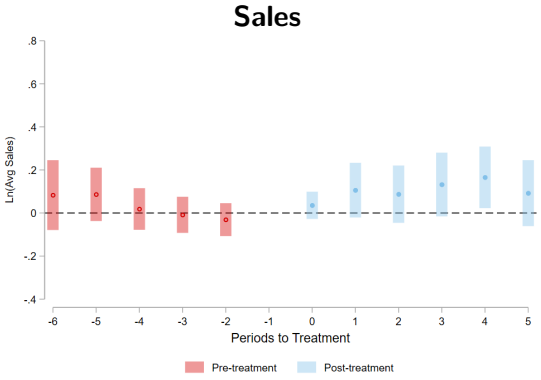


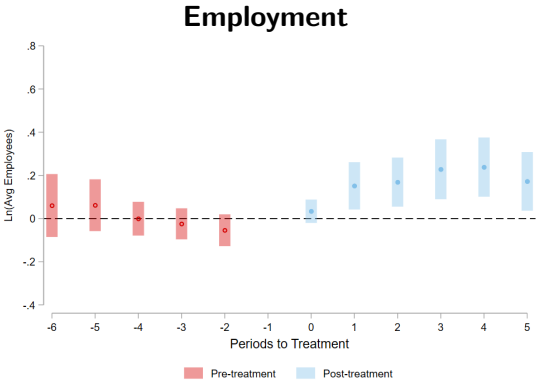


Robustness

Balanced event-time estimates preserve the main post-treatment employment pattern.

Backup: balanced event-time window (-5,+5), excluding NAICS 71–72





Robustness

Effects outside gaming and hospitality remain more modest, especially for sales.

Backup: Baseline balance (1990): Ever-treated (ET) vs. never-treated (NT) reservations

	Mean		Diff.	SMD	p-val.	Obs.	
	ET	NT				ET	NT
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Baseline log sales (1990)	13.186	13.019	0.167	0.133	0.304	131	108
Baseline log emp. (1990)	2.154	2.183	-0.029	-0.027	0.838	131	108
Emp. share: Retail (1990)	0.103	0.098	0.005	0.025	0.848	131	108
Emp. share: Hospitality (1990)	0.148	0.163	-0.015	-0.055	0.670	131	108
Emp. share: Manufacturing (1990)	0.084	0.045	0.040	0.255	0.055	131	108
Poverty rate (1990)	0.332	0.320	0.011	0.065	0.639	114	94
Pretrend in log emp. (1991–1990)	0.005	0.034	-0.029	-0.148	0.255	128	108
Pretrend in log sales (1991–1990)	0.071	0.090	-0.019	-0.055	0.679	128	108

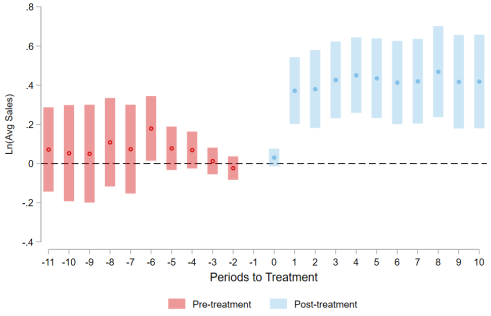
Notes: The table reports baseline (1990) means by group, the mean difference, a two-sided p-value from a difference-in-means t-test, and the standardized mean difference (SMD). Reported *Obs*. equals the number of reservations contributing to each row (pairwise-complete observations), so sample sizes can vary across rows. Poverty rates in 1990 are unavailable for some reservations in the NHGIS/IPUMS tabulations. For pretrend measures (1991–1990), reservations first treated in 1991 are excluded to avoid contamination of 1991 outcomes.

Backup: Baseline balance (1990): Early-treated (ET, 1991–1996) vs. never-treated (NT) reservations

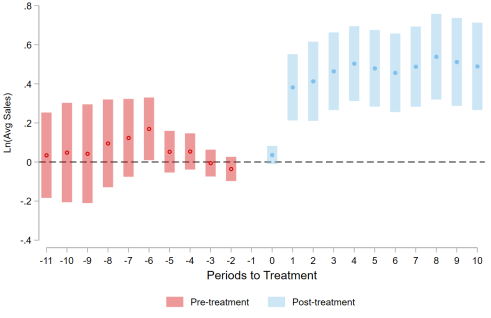
	Mean		Diff.	SMD	p-val.	Obs.	
	ET	NT				ET	NT
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Baseline log sales (1990)	13.295	13.019	0.277	0.213	0.177	67	108
Baseline log emp. (1990)	2.277	2.183	0.095	0.086	0.579	67	108
Emp. share: Retail (1990)	0.096	0.098	-0.003	-0.014	0.927	67	108
Emp. share: Hospitality (1990)	0.155	0.163	-0.008	-0.031	0.846	67	108
Emp. share: Manufacturing (1990)	0.083	0.045	0.039	0.251	0.095	67	108
Poverty rate (1990)	0.341	0.320	0.021	0.120	0.473	57	94
Pretrend in log emp. (1991–1990)	0.011	0.034	-0.023	-0.114	0.473	64	108
Pretrend in log sales (1991–1990)	0.027	0.090	-0.063	-0.206	0.194	64	108

Notes: The table reports baseline (1990) means by group, the mean difference, a two-sided p-value from a difference-in-means t-test, and the standardized mean difference (SMD). Reported *Obs*. equals the number of reservations contributing to each row (pairwise-complete observations), so sample sizes can vary across rows. Poverty rates in 1990 are unavailable for some reservations in the NHGIS/IPUMS tabulations. For pretrend measures (1991–1990), reservations first treated in 1991 are excluded to avoid contamination of 1991 outcomes.

Backup: conditional specifications, sales

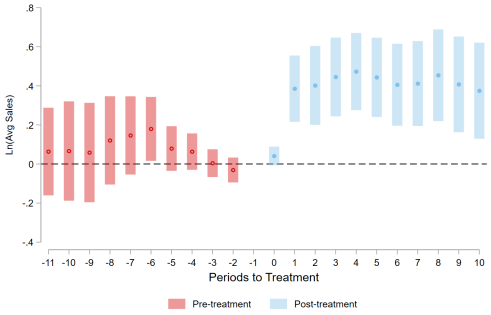


Main

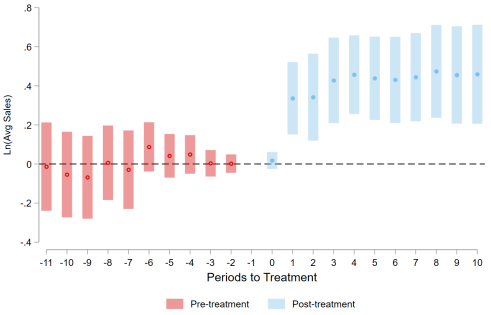


Baseline log level

Backup: conditional specifications, sales

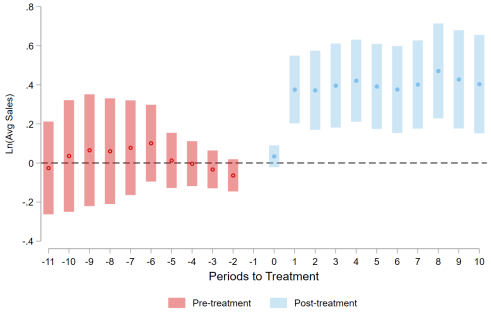


Industry shares

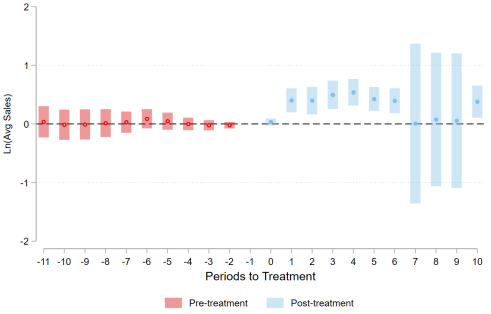


Poverty rate

Backup: conditional specifications, sales



Pre-trends



All covariates

Ongoing work

Current revision priorities

- Separate direct casino effects from broader reservation business spillovers.
- Add total sales and total employment outcomes to measure aggregate reservation business expansion.
- Expand heterogeneity analyses by casino size and reservation size.
- Add spatial spillover analysis using distance buffers around reservations and casino locations.
- Strengthen robustness analyses and sample-exclusion checks.